

Reasoning:

In scientific inquiry, there is often disagreement, meaning the results of one study do not align with another. This lesson is designed to give students a much richer grasp of why scientists (and people) disagree, and how these disagreements can be resolved. Understanding ways to resolve disagreements is critical both for lay adults trying to make sense of science in the real world, and to understanding why scientists disagree and how they resolve disagreement. Students analyze scientific disagreements and work to come to consensus with their peers prior to analyzing the data that was collected by the class. This lesson allows students to understand that consensus does not mean agreeing on a single scientists' conclusion; conditional conclusions (i.e. one recommendation might work better for one community and another recommendation might work for another) are likely when attempting to analyze complex real-world scenarios. At the conclusion of this lesson, students should have an expanded understanding of “coming to consensus” in science.

Slide	Notes
DO NOW! LESSON 5 When investigating similar questions, scientists often try to resolve disagreements. Why would they not just “agree to disagree”? Instructions: Turn and talk to a partner about your thoughts, and type them in our “Thoughts & Reflections” sheet (in the tab with your name on it).	Suggested: Have a class discussion on this question after students have thought of initial ideas. Scientists have to come to conclusions that align all of the valid evidence that has been collected. This will likely lead to conditional conclusions to explain a single complex phenomenon.
Two or three groups each investigated the same mitigation factor. What if you got different results? Should you consider only your own results?	Pose this question and elicit students’ responses. Hopefully, students will come to realize that they cannot dismiss others’ evidence, and that their ideal of fitting all the good evidence means that if they determine that if each team conducted a properly controlled experiment, they need to work out an answer that fits ALL the evidence. If students are struggling to address this question, consider making the connection to Amazon reviews. Would they rather buy a product with one five star review, or a product with 100 reviews that averaged 4.5 stars? It’s not anchored in science specifically, but usually this provokes a good conversation

WHY DO SCIENTISTS DISAGREE SOMETIMES?

In scientific inquiry, there is often disagreement. For example, some scientists may say that cleaning surfaces is highly effective for avoiding the spread of a disease. Others may say that cleaning surfaces is only moderately effective, and others may say that it is not at all effective.

Why do you think scientists disagree on these kinds of issues?

Topics that might be addressed during a class discussion (if they are not, ask students question to elevate these ideas):

- Different studies may point in different directions.
- There hasn't been enough research yet to converge on an answer.
- There might be different details of experiments. For example, one study might study surface cleaning over 1 week, and another studies it over 1 month; maybe those details make a difference.
- There might be different methods. Maybe some studies turn out to be flawed, and then you find out that this is why some studies yielded different results.
- If there is not enough data yet, different scientists might interpret the results of studies in different ways.

How can scientists resolve these disagreements? (In other words, how can they come to agreement?)

Some key ideas relevant to DeTECT learning outcomes that could be drawn out if they arise, or be suggested if they don't:

- Look at all the studies, not just some of them. You can see patterns better if you look at ALL of the evidence.
- Carefully examine studies to see how they are different.
- Maybe some have flawed methodologies and can be dismissed.
- Maybe there are differences in the details of the studies (duration of the study, the exact details of the conditions, the nature of the control group), that can help you decide.
- After careful examination of the differences among studies, gather more data to try to resolve differences.

Note that resolution strategies are connected to the reasons for disagreement. This connection is often not obvious!

Close by noting that next they are going to learn some of the ways that scientists can resolve disagreements.

ANALYZING DISAGREEMENTS

Lead students into the next activity, where the class will analyze two disagreements. Ideally, students will first analyze the disagreements in groups, and then bring their thoughts to a class discussion.

Skipping the smaller group analyses and moving directly to a class discussion might be a way to save time if needed.

DISAGREEMENT 1: WHY DO THEY HAVE DIFFERENT CONCLUSIONS?

You can find those groups' experiments in the Disagreement 1 tab in your Experiment Report.

1. What's different about these groups' conclusions?
2. Why might they disagree about the conclusion?
3. What conclusion would fit all the high-quality evidence?

When discussing Disagreement 1 it would be helpful to discuss the sociological aspects that affect which mitigation strategies are chosen. Some people have to go to work even with lockdowns, and the longer the lockdown goes, the more economically challenging it is to some people who are not allowed to work during a lockdown. This disagreement involves different but reasonable choices of what the most meaningful experiment is given real-world constraints.

Answers to these 3 questions:

1. The time frame during which the mitigation strategy was tested.
2. The students disagree because they used different methods. Specifically, one group examined lockdowns over 20 days, and one group examined lockdowns over 60 days. These are not methodological flaws--each group has some reasonable grounds for choosing what they choose.
3. One way to resolve this disagreement is to say that the effect of lockdowns depends on how long you run the lockdowns for. The effects are small over 20 days, but large over 60 days.

Emphasize that **if** there is a serious flaw with one study, then that study could be dismissed. (Examples are a confounded study, or a study with only 1 trial per condition.) Hopefully, the fact that they all got feedback on their experiments and had a chance to revise them will mean that their experiments aren't flawed and so won't have to be dismissed. But if they are flawed, they should be dismissed.)

Emphasize the importance of ample evidence: To explain all the evidence, we have to dig in and work

	out these differences to understand what is going on. Neither group can get to the best answer by themselves. They need to share their evidence to find out what the best answer is. They need to use all the evidence.
DISAGREEMENT 2: WHY DO THEY HAVE DIFFERENT CONCLUSIONS? You can find those groups' experiments in the Disagreement 2 tab in your Experiment Report. 1. What's different about these groups' conclusions? 2. Why might they disagree about the conclusion? 3. What conclusion would fit all the high-quality evidence?	When discussing Disagreement 2, it would be helpful to discuss the sociological aspects that affect which experiments are conducted. You might want to find the effects of a variable in isolation. Or you might want to find out if your community should implement it on top of other mitigation strategies that they are more willing to use. These are both legitimate research strategies, and which strategy you choose may depend on what question you think is most important for your community to answer. Answers to these 3 questions: 1. Students might say they disagree, or that they don't really disagree because they did the experiments differently. (They only disagree on the surface.) 2. The students disagree because they used different methods. One group examined lockdowns by themselves, as the only mitigation strategy. The other group examined lockdowns on top of two other mitigation strategies--handwashing and vaccinations. These differences do not involve methodological flaws--each group has some reasonable grounds for choosing what they choose, and each experiment is properly controlled. 3. One way to resolve this disagreement is to conclude that the effect of lockdowns depends on whether they are used by themselves, or with other mitigation strategies. As the sole mitigation strategy, they have a very large effect. When added to two other strategies (handwashing and vaccinations), they have a more modest effect. It is good to discuss that further research can examine these questions further, and provide support for this. Ask students what experiment they could run to test their new conclusion. Emphasize the importance of ample evidence: To explain all the evidence, we have to dig in and work out these differences to understand what is going on better. Neither group can get to the best answer by

	themselves. They need to share their evidence to find out what the best answer is. They need to use all the evidence. You can also discuss with students lockdowns might not be chosen even if they have some positive effects. If lockdowns save only a few lives, there might be reasons not to choose lockdowns. There is risk to life as a result of stopping work, as well.
	Next, students will use the strategies they have learned to analyze their own results. Have all students who investigated the same mitigating factor get into groups. Their overall goal (to be achieved on the next slide) is to develop a conclusion that fits all their high-quality evidence. Using their schemas just as in the three examples, they will compare their results, and determine if their results are the same or not. They should ask themselves: does my conclusion fit all the evidence, not only my evidence? In their first steps, they carefully compare their results and their conclusions. They should be very precise, and not ignore small differences. They want to precisely identify whether their own conclusion fits other groups' evidence. After students go through the questions on the Review of Research page, you can highlight one interesting group's work (or the work of more than one group, if you have time), to show how they have analyzed their differences. • Emphasize the power of these ways of analyzing and resolving disagreements. • Emphasize that scientists use these same approaches. Now students build on their analysis so far to develop a conclusion that accounts for all the high-quality data. They can exclude data from low-quality studies. This will mainly mean experiments with just one trial per condition, and experiments that are confounded.